

VOLATILITY PREDICTION MODEL

Introduction & Motivation

"Volatility is the price of admission for superior returns." - Morgan Housel
Developed a **Machine Learning model** to predict market volatility, enabling informed investment decisions amidst recent market fluctuations.

Data & Feature Engineering

- Utilized historical SPY data (Close, High, Low, Open, Volume) from June 2023 to July 2023.
- Engineered features: Returns, Rolling Volatility, ATR, Skewness, Kurtosis, Bollinger Bands, and Average Volume.
- Data window: 14 days for rolling calculations.

Model Training

Implemented a **Random Forest Classifier** (200 estimators, balanced weights) to predict volatility levels.
It was not overfitting; our problem was just too easy for the model.

Results & Insights

Model performance evaluated using a confusion matrix and accuracy scores.
Correlation matrix revealed key feature relationships, guiding feature selection.